

Expanding, Factorising & Solving Equations

Quick reference sheet · GCSE Maths, Algebra · keep handy for revision

1. Expanding single brackets

$a(b + c) = ab + ac$ — multiply everything inside by what is outside. A negative multiplier flips every sign inside.

$$3(x + 4) = 3x + 12 \quad \cdot \quad -2(x + 6) = -2x - 12 \quad \cdot \quad -(3x - 5) = -3x + 5$$

2. Expanding double brackets (FOIL)

Two brackets multiplied together, e.g. $(x + a)(x + b)$, need **every term in the first bracket** multiplied by **every term in the second bracket** — that is 4 separate multiplications in total. **FOIL** is just a way to remember all 4 of them:

First — multiply the first term in each bracket

Outer — multiply the two outer terms (first bracket's first term, second bracket's last term)

Inner — multiply the two inner terms (first bracket's last term, second bracket's first term)

Last — multiply the last term in each bracket

Then add all 4 results together and simplify by collecting like terms.

Worked example, step by step: $(x + 3)(x + 5)$

F: $x \times x = x^2$

O: $x \times 5 = 5x$

I: $3 \times x = 3x$

L: $3 \times 5 = 15$

Add them together: $x^2 + 5x + 3x + 15 \rightarrow$ collect the two x-terms $\rightarrow x^2 + 8x + 15$

Watch the signs when a bracket has a minus in it — e.g. in $(x - 2)(x + 7)$, the “3” slot above becomes -2 , so Inner is $-2 \times x = -2x$, and Last is $-2 \times 7 = -14$.

Brackets	Expanded & simplified
$(x + 2)(x + 6)$	$x^2 + 8x + 12$
$(x - 2)(x + 7)$	$x^2 + 5x - 14$
$(x - 4)(x - 3)$	$x^2 - 7x + 12$

3. Factorising

Factorising is the **reverse** of expanding — you start with an expanded expression and write it as brackets multiplied together.

3a. Common factor

Find the highest common factor (HCF) of every term — the biggest number (and/or letter) that divides into all of them — then write it outside a bracket containing what is left of each term.

Worked example: $6x + 9$

HCF of 6 and 9 is 3 (both divide by 3, and nothing bigger does)

$6x \div 3 = 2x$, $9 \div 3 = 3 \rightarrow$ write these inside the bracket

Answer: **$3(2x + 3)$** — check by expanding back: $3 \times 2x + 3 \times 3 = 6x + 9$ ✓

Another: $10x^2 - 15x \rightarrow$ HCF is $5x \rightarrow$ **$5x(2x - 3)$**

3b. Factorising a quadratic $x^2 + bx + c$

A quadratic like $x^2 + bx + c$ factorises into two brackets, $(x + p)(x + q)$. Because expanding those brackets gives $x^2 + (p+q)x + pq$ (compare to FOIL above), you need to find two numbers p and q that:

multiply together to give c (the number on its own), **and**

add together to give b (the number of x 's)

Worked example, step by step: $x^2 + 7x + 12$

Need two numbers that multiply to 12 and add to 7

Try factor pairs of 12: (1,12) add to 13 ✗ (2,6) add to 8 ✗ (3,4) add to 7 ✓

Answer: **$(x + 3)(x + 4)$** — check by expanding back (FOIL): $x^2 + 4x + 3x + 12 = x^2 + 7x + 12$ ✓

If c is negative, one of your two numbers must be negative. If b is also negative, the bigger number (by size) must be the negative one.

$x^2 + 2x - 15$ — need two numbers multiplying to -15 , adding to 2 $\rightarrow$ 5 and $-3 \rightarrow$ **$(x + 5)(x - 3)$**

$x^2 - 9x + 20$ — need two numbers multiplying to 20, adding to $-9 \rightarrow$ -4 and $-5 \rightarrow$ **$(x - 4)(x - 5)$**

4. Solving linear equations

Balance method: do the same operation to both sides until x is alone. Expand any brackets first; collect x terms onto one side if x appears twice.

$$3x + 7 = 22 \rightarrow 3x = 15 \rightarrow \mathbf{x = 5}$$

$$2(x + 3) = 16 \rightarrow 2x + 6 = 16 \rightarrow 2x = 10 \rightarrow \mathbf{x = 5}$$

$$5x - 3 = 2x + 9 \rightarrow 3x - 3 = 9 \rightarrow 3x = 12 \rightarrow \mathbf{x = 4}$$

5. Check your work

After factorising, expand your answer back out — it should match the original expression. After solving an equation, substitute your value of x back in to check both sides are equal.